The same pattern works for matrices of any size. This kind of matrix multiplication is sometimes called the *Kronecker product*, a term that only applies to matrices—it's the matrix version of the tensor product. The Kronecker product of two 2 × 2 matrices is a 4 × 4 matrix, and the pattern is similar for matrices of arbitrary size. In general, the Kronecker product of an $m \times n$ matrix and a $p \times q$ matrix is an $mp \times nq$ matrix.

All of this applies perfectly well to column and row vectors, which are just specialized matrices. The tensor product of two 2 × 1 column vectors is a 4 × 1 column vector. If $a$ and $b$ are 2 × 1 column vectors, their tensor product looks like this:

$$\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \otimes \begin{pmatrix} b_{11} \\ b_{21} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} \\ a_{11}b_{21} \\ a_{21}b_{11} \\ a_{21}b_{21} \end{pmatrix}. \tag{7.8}$$

Let's see how this works out for Alice and Bob. First, we'll construct the four tensor product basis vectors, using $|u\rangle$ and $|d\rangle$ as building blocks. Recall Eqs. 2.11 and 2.12 from Lecture 2,

$$|u\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|d\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

If we plug the appropriate combinations of $|u\rangle$ and $|d\rangle$ into Eq. 7.8, our four 4 × 1 column vectors are